

LORRIO — APP & TECHNICAL ARCHITECTURE SPECS

Full Technical Specifications, Component Architecture & Deployment Details

1. Technology Stack Overview

- **Frontend Framework:** Next.js 14 (App Router) with React 18 & TypeScript 5.5+
- **Styling & Design System:** Tailwind CSS with custom Laterite Earth theme palette
- **State Engine:** React Context API (``LorrioContext.tsx``) with real-time multi-role state transitions
- **Deployment Infrastructure:** Netlify Static Export (``out/``) with ``netlify.toml`` CI/CD pipeline
- **Source Control:** GitHub repository (``alishanea/lorrio``)

2. Built Software Components

1. **Role Landing Page (``RoleLandingPage.tsx``):** Interactive entry portal presenting 3 role cards: Customer (I Need Material), Driver (I Have A Lorry), and Quarry Supplier (I Own A Quarry).
2. **Mobile App Shell (``MobileAppShell.tsx``):** Native mobile layout with top location bar, profile badge, and fixed bottom navigation bar (Home, Loads, Orders, Alerts).
3. **Location Picker Map (``LocationPickerMap.tsx``):** Interactive map pin selection widget with GPS coordinate output (``11.6094° N, 76.0827° E``).
4. **Price & Freight Estimator (``PriceEstimatorWidget.tsx``):** Real-time quote calculator with quantity slider (200-1,500 stones) and instant cost itemization.
5. **WhatsApp API Integration (``generateWhatsAppUrl``):** Pre-filled WhatsApp link generator containing order ID, lorry number, driver phone, ETA, total amount, and live tracking link.
6. **Driver ETA Setter (``DriverApp.tsx``):** Modal allowing lorry drivers to update arrival ETAs in real time.
7. **Notification Center (``NotificationCenter.tsx``):** Real-time alert feed for customers and drivers.
8. **Live GPS Tracking Map (``LiveGpsTrackerMap.tsx``):** Interactive map simulation animating lorry movement along Kannur-Wayanad ghat pass.